

■ IEEE Brain Tumor Project

Complete Reference Document

Super-Resolution Enhanced Brain Tumor Classification Using EfficientNetB3 on Degraded MRI Images

Conference	National IEEE
Dataset	Kaggle Brain Tumor MRI (7023 images, 4 classes)
SR Models	SRCNN vs EDSR (Comparative Study)
Classifier	EfficientNetB3 (Transfer Learning)
Localization	YOLOv8
Explainability	Grad-CAM
Frontend	React + Vite + Three.js → Vercel
Backend	Express.js + MongoDB → Railway
AI Server	FastAPI → HuggingFace Spaces
Submission	August 2025

1. Team Division

Member	Role	Responsibilities
Member 1 (Team Lead)	AI Core	SRCNN training, EfficientNetB3, Degradation experiments, PSNR/SSIM, Training metrics
Member 2 (You)	SR Pipeline + Detection	EDSR implementation, YOLOv8, Grad-CAM, FastAPI integration, 3D coord mapping
Member 3	Frontend	React + Vite UI, Three.js 3D Brain, API connections, Vercel deploy
Member 4	Auth + Paper	Express.js, MongoDB, JWT auth, Literature review, IEEE formatting

2. Complete Pipeline

Step	Module	Input → Output	Owner
1	Degradation	Original MRI → LR_Testing	M2 (You) ■ Done
2	SRCNN	LR_Testing → SRCNN_Testing	M2 (You) ■ Done
3	EDSR	LR_Testing → EDSR_Testing	M2 (You) ■ Next
4	EfficientNetB3	All datasets → Accuracy tables	M1 ■ Pending
5	YOLOv8	Enhanced MRI → Bounding boxes	M2 (You) ■ Pending
6	Grad-CAM	Model + Image → Heatmap	M1 ■ Pending
7	FastAPI	All models → API endpoint	M2 (You) ■ Pending
8	React + Three.js	API results → 3D visualization	M3 ■ Pending
9	Express + MongoDB	Doctor login + trial count	M4 ■ Pending
10	Deploy	Vercel + Railway + HuggingFace	All ■ Pending

3. Current Results (SRCNN)

SRCNN vs LR_Testing Metrics:

Method	PSNR (dB)	SSIM	Status
LR (Degraded)	Baseline	Baseline	Reference
SRCNN Enhanced	29.82	0.918	■ Complete
EDSR Enhanced	Pending	Pending	■ Next Step

Result Analysis:

- PSNR 29.82 dB = Good range (25-35 dB acceptable for medical imaging)
- SSIM 0.918 = Excellent (>0.9 = excellent structural similarity)
- EDSR expected to give PSNR ~31-33 dB and SSIM ~0.92-0.94
- PSNR Paradox possible: EDSR higher PSNR but SRCNN higher classification accuracy — this is a valid paper finding!

4. Pending Implementation Tasks

Your Tasks (Member 2):

Priority	Task	Est. Time
1 - Immediate	EDSR model train + EDSR_Testing generate	1-2 days
2 - After EDSR	PSNR/SSIM comparison (SRCNN vs EDSR)	1 hour
3 - After M1 trains	YOLOv8 annotation + training	1 week
4 - After YOLOv8	Grad-CAM implementation + IoU	2-3 days
5 - Integration	FastAPI — all models in one endpoint	3-4 days
6 - Integration	3D coordinate mapping YOLO → Three.js	1 day

Member 1 Tasks:

Priority	Task	Est. Time
1	EfficientNetB3 train on LR_Testing	2-3 hours (Colab)
2	EfficientNetB3 train on SRCNN_Testing	2-3 hours (Colab)
3	EfficientNetB3 train on EDSR_Testing (after you generate)	2-3 hours
4	Classification report + confusion matrix all datasets	1 day
5	Grad-CAM figures — with/without SR comparison	2 days

5. IEEE Paper — What to Include

Tables Required:

Table	Content	Data Source
Table 1	SR Quality: LR vs SRCNN vs EDSR (PSNR/SSIM)	Your SR experiments
Table 2	Classification: LR vs SRCNN vs EDSR accuracy per class	Member 1 experiments
Table 3	Existing methods comparison (VGG16, ResNet, DenseNet, EfficientNetB3, Ours)	Need to compare your results
Table 4	Degradation levels study (Mild/Medium/Severe)	Ablation experiments

Figures Required:

Figure	Content
Figure 1	System pipeline diagram (LR → SR → EfficientNetB3 → YOLOv8 → Grad-CAM → 3D)
Figure 2	Visual comparison: Original LR Degraded SRCNN EDSR (side by side)
Figure 3	Grad-CAM: Without SR (scattered) vs With SR (focused on tumor)
Figure 4	Training loss curves for SRCNN and EDSR
Figure 5	Confusion matrix for EfficientNetB3 on best SR method

Paper Sections:

Section	Content	Status
Abstract	Problem + Method + Dataset + Results numbers	■ Draft ready
Introduction	Brain tumor importance + DL gap + Your contributions	■ August
Related Work	SRCNN, EDSR, EfficientNet, YOLOv8 papers cite	■ August
Methodology	Degradation + SRCNN + EDSR + B3 + YOLOv8 + Grad-CAM	■ August
Experiments	Dataset + Training settings + All tables	■ August
Results	All figures + Analysis + Discussion	■ August
Conclusion	Summary + Limitations + Future work	■ August

6. Final Tech Stack

Layer	Technology	Deploy	Purpose
AI Models	PyTorch	-	Framework
SR	SRCNN + EDSR	HuggingFace	MRI enhancement
Classifier	EfficientNetB3	HuggingFace	4-class tumor classification
Detector	YOLOv8	HuggingFace	Tumor localization
Explainability	Grad-CAM	HuggingFace	Why model decided
AI Server	FastAPI	HuggingFace	Model serving API
Auth Gateway	Express.js	Railway	Doctor login + trials
Database	MongoDB Atlas	Cloud	Doctor accounts
Frontend	React + Vite	Vercel	Doctor UI
3D Brain	Three.js + R3F	Vercel	3D tumor visualization
PDF Report	FPDF2	HuggingFace	Clinical report

Communication Flow:

Doctor Browser (Vercel) → POST MRI → Express.js (Railway) → Auth check + Trial count → FastAPI (HuggingFace) → SRCNN/EDSR + EfficientNetB3 + YOLOv8 + Grad-CAM → JSON response → Express → React displays results + Three.js 3D brain

7. Key Technical Risks

Risk	Probability	Solution
EfficientNetB3 overfit (3 class small data)	High	Augmentation + Dropout 0.5 + Label smoothing
EDSR hallucinations (fake tumor features)	Medium	Use MSE loss only, NOT perceptual loss
PSNR Paradox (EDSR higher PSNR, lower accuracy)	Low	Report as novel finding in paper!
Class imbalance (No Tumor = fewer images)	High	Class weights in CrossEntropyLoss
Grad-CAM instability (different maps each run)	Low	Use SmoothGradCAMpp instead
Colab disconnect during training	High	Save checkpoint every 10 epochs to Drive
YOLOv8 annotation time (1311 images)	High	Labellmg tool, divide among team members
Team members not contributing	High	Weekly deadlines, simple tasks, starter code

8. Timeline

Month	You (M2)	Member 1	Members 3+4
March (Now)	■ LR_Testing done ■ SRCNN done ■ EDSR next	■ EfficientNetB3 train on all datasets	■ React setup ■ Express setup
April	■ YOLOv8 training ■ Grad-CAM ■ FastAPI setup	■ Accuracy tables ■ Grad-CAM figures ■ Confusion matrix	■ React UI pages ■ MongoDB setup
May	■ FastAPI complete ■ All models integrated ■ API testing	■ Ablation study ■ Results analysis	■ Three.js 3D brain ■ JWT auth
June	■ Integration ■ End-to-end testing ■ Bug fixes	■ Final experiments ■ Figures finalize	■ Deploy ■ Integration test
July	■ Paper writing (Methodology)	■ Paper writing (Results + Experiments)	■ Paper writing (Lit review, Format)
August	■ IEEE Submit!	■ IEEE Submit!	■ IEEE Submit!

9. Prerequisites — What to Learn

Member	Topics	Resources
M1 + M2 (DL Team)	Python, PyTorch, OpenCV, SRCNN, EDSR, EfficientNetB3, FastAPI, Google Colab	FastAPI, Google Colab, freeCodeCamp
M3 (Frontend)	React + Vite, Three.js, @react-three/fiber, @react-three/drei, Tailwind CSS	react-three.js, docs.pmnd.rs, YouTube Traversy
M4 (Auth)	Node.js, Express.js, JWT, bcrypt, MongoDB, Mongoose	YouTube Net Ninja, mongoosejs.com
All Members	Git + GitHub, VS Code, Branch workflow, Pull requests	youtu.be/RGOj5yH7evk

10. All Locked Decisions

Decision	Value
Paper Title	Super-Resolution-Enhanced Brain Tumor Classification Using EfficientNetB3 on Degraded MRI Images
SR Models	SRCNN vs EDSR (comparative study inside paper)
Classifier	EfficientNetB3 (300x300 input, 12M params, nobody in reference papers used B3)
Dataset	Kaggle Brain Tumor MRI — Nickparvar (7023 images, 4 classes)
Classes	Glioma, Meningioma, Pituitary, No Tumor
Framework	PyTorch (not TensorFlow)
Degradation	Downsampling + Gaussian blur (simulate real clinical conditions)
Metrics SR	PSNR + SSIM
Metrics Classify	Accuracy + Precision + Recall + F1 + Confusion Matrix
Metrics Localize	YOLOv8 mAP50 + Grad-CAM IoU
AI Deploy	FastAPI on HuggingFace Spaces
Auth Deploy	Express.js + MongoDB on Railway
Frontend Deploy	React + Vite + Three.js on Vercel
Auth Features	Doctor login, JWT tokens, 5 free trials limit
Conference	National IEEE (August 2025 submit)

11. 3D Coordinate Mapping Formula

YOLO pixel coordinates → Three.js 3D brain space:

Variable	Formula	Description
brain_x	$(cx / img_size - 0.5) \times 160$	Left-Right position
brain_y	$(cy / img_size - 0.5) \times 130$	Front-Back position
brain_z	0 (approximate)	Depth (center slice)
tumor_size	$((x2-x1)+(y2-y1)) / 2 / img_size \times 40$	Sphere radius in Three.js

- Note: Paper writing starts August. Focus on implementation first. EDSR is next immediate step.
- Germany MS Goal: IEEE paper + strong implementation = powerful SOP + CV for MS applications.